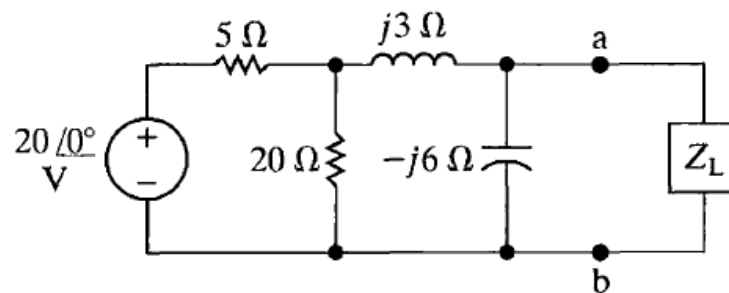
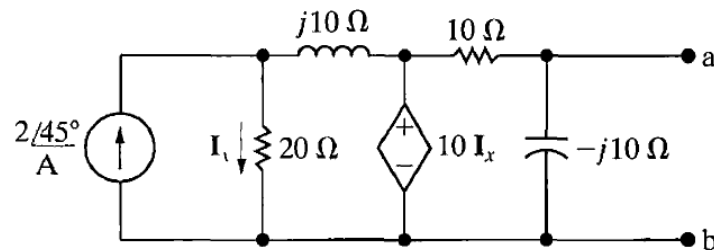


- For the circuit shown in Fig., determine the impedance  $Z_L$  that results in maximum average power transferred to  $Z_L$ .  
What is the maximum average power transferred to the load impedance determined in (a)?



- Find the Thevenin equivalent with respect to terminals a,b in the circuit shown.



Solution

- $\mathbf{V_{Th}} = 11.52 - j15.36 \text{ V.} \quad \mathbf{Z_{Th}} = 5.76 - j1.68 \text{ } \Omega. \quad \mathbf{Z_L} = 5.76 + j1.68 \text{ } \Omega.$

$$P = I_{\text{eff}}^2(5.76) = 8 \text{ W}$$

- $\mathbf{V_{Th}} = \mathbf{V_{ab}} = 10 \angle 45^\circ \text{ V.}$   
 $\mathbf{Z_{Th}} = 5 - j5 \text{ } \Omega.$